CartPilot

Inventory Management System

User Guide

Hexabyte Team | CSCI2040U | Ontario Tech University

1. Introduction

CartPilot is a desktop inventory management system designed for grocery store staff. It allows teams to manage product inventory, track activity, and control access through role-based permissions. The system runs on Windows and connects to a MySQL database to store all data persistently. This guide explains how to use CartPilot from the perspective of each user role — Owner, Manager, and Employee.

2. Getting Started

2.1 Launching the Application

Double-click the application or run the following command from the project folder:

```
build.bat
```

The CartPilot login screen will appear.

2.2 First Time Setup

If this is the first time CartPilot has been launched on this machine, you will see a dialog asking you to create an Owner account. This is the administrator account for the system with full access to all features.

1. Enter a username for the Owner account
2. Enter a password (minimum 4 characters)
3. Click '*OK*'

⚠ **Note:** This step only appears once. If you click Cancel the application will close.

2.3 Logging In

1. Enter your username in the Username field
2. Enter your password in the Password field
3. Click '*Login*' or press '*Enter*'

You will be taken to the dashboard for your role — Owner/Manager Dashboard or Employee Dashboard.

3. User Roles

CartPilot has three user roles, each with different levels of access:

Role	Description	Access Level
Owner	Full administrator — can do everything including managing all users	Full
Manager	Can manage inventory and add/remove employee accounts	High
Employee	Can view inventory, search, and update quantities only	Limited

4. Owner & Manager Dashboard

After logging in as an Owner or Manager you will see the main dashboard with the inventory table and action buttons at the bottom.

4.1 Viewing Inventory

The inventory table displays all items in the database with the following columns:

- ID — unique identifier for the item
- Name — product name
- Price — current price in dollars
- Quantity — current stock level
- Origin — origin of the item

💡 **Tip:** Click Refresh to reload the latest data from the database at any time.

4.2 Searching Inventory

Type in the Search bar above the inventory table to filter items by name in real time. The table updates as you type. Clear the search field to show all items again.

4.3 Adding an Item

1. Click the '*Add Item*' button
2. Fill in all fields: Item ID, Name, Price, Quantity, and Origin
3. Click '*OK*' to save

⚠ **Note:** Item ID must be a positive whole number and must not already exist. Price and Quantity cannot be negative.

4.4 Removing an Item

1. Click the row of the item you want to remove in the table
2. Click the '*Remove Item*' button or press '*R*' on your keyboard
3. Confirm the deletion by clicking '*Yes*'

4.5 Updating Price

1. Click the row of the item in the table
2. Click '*Update Price*' or press '*P*' on your keyboard
3. Enter the new price and click '*OK*'

4.6 Updating Quantity

1. Click the row of the item in the table
2. Click '*Update Quantity*' or press '*Q*' on your keyboard
3. Enter the new quantity and click '*OK*'

💡 **Tip:** You can also press = to increase quantity by 1 or - to decrease by 1 on a selected row.

4.7 Viewing the Activity Log

Click '*Clear Log*' to completely clear all log entries.

4.8 Clearing the Activity Log

Click '*View Log*' to open the activity log. The log shows every action taken in the system with the username and timestamp of who performed it, ordered from most recent to oldest.

4.9 Keyboard Shortcuts

Key	Action
=	Increase the selected item quantity by 1
-	Decrease the selected item quantity by 1
R	Remove the selected item
P	Update price of the selected item
Q	Update quantity of the selected item

5. Managing Users

User management is available from the Settings menu in the top menu bar of the Owner/Manager Dashboard.

5.1 Adding a New User

1. Click '*Settings*' in the menu bar
2. Click '*Add New User'*
3. Enter a username and password (minimum 4 characters)
4. Select a role from the dropdown
5. Click '*OK*'

⚠ **Note:** Managers can only add Employee accounts. Owners can add any role. Usernames must be unique.

5.2 Removing a User

1. Click '*Settings*' in the menu bar
2. Click '*Remove User*'
3. Enter the username of the account to remove
4. Click '*OK*'

⚠ **Note:** Managers can only remove Employee accounts. Owners can remove any account.

6. Employee Dashboard

Employees have a simplified dashboard showing the inventory table with limited actions.

6.1 Viewing & Searching Inventory

The inventory table loads automatically. Use the Search bar to filter by item name in real time. Click '*Refresh Inventory*' to reload the latest data.

6.2 Updating Quantity

1. Click the row of the item in the table
2. Click '*Update Quantity*' or press '*Q*'
3. Enter the new quantity and click '*OK*'

💡 **Tip:** Press = to increase quantity by 1 or - to decrease by 1 on a selected row.

7. Logging Out

Click the '*Logout*' button at the bottom of the dashboard to return to the login screen. Your session data will not be saved — you will need to log in again next time, however any inventory changes will be saved.

8. Troubleshooting

Problem	Solution
Application won't launch	Ensure MySQL Server is running before launching CartPilot
Database connection failed	Check that your credentials in InventorySystem.java match your MySQL setup
Login says Invalid username or password	Check caps lock and re-enter credentials carefully
Item won't add — ID already exists	Use a different Item ID — each ID must be unique
Cannot remove a user	Managers can only remove Employee accounts — contact an Owner for other roles
Table appears empty after refresh	Check that MySQL is running and the inventory table exists

9. Security Notes

- Passwords are hashed using SHA-256 before being stored in the database — they cannot be retrieved in plaintext. This protects passwords even if the database was breached.
- Only Owners can register new Owner or Manager accounts.
- All actions are recorded in the activity log with the username and timestamp.
- There is no password reset feature — contact your Owner to have your account removed and recreated.

10. Contact & Support

CartPilot was developed by the Hexabyte team for their customer, Mr. David, to manage his grocery store. For technical issues contact your system administrator or the development team.

GitHub Repository: https://github.com/EmmanuelRodrigues791/Hexabyte-INC.git